Coupling the Point Model to the Growth Field

Key points of the meeting of 20 October 2026

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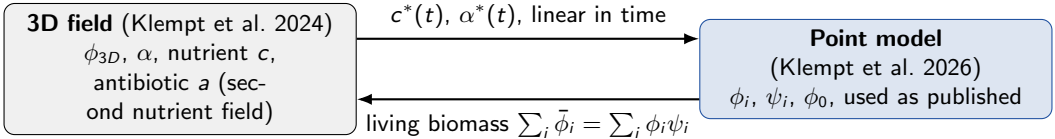
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20 October 2026

What changes

- Point model (Klempt et al. 2026) used as published
- No scaling to the field ($\phi_0 = 1 - \phi_{3D}$ dropped)
- Field $\rightarrow$ point model: c^*, α^* from 3D fields
- Point model $\rightarrow$ field: only living biomass consumes
- Eq. 34 with front term, Eq. 36 unchanged

The loop at one Gauss point



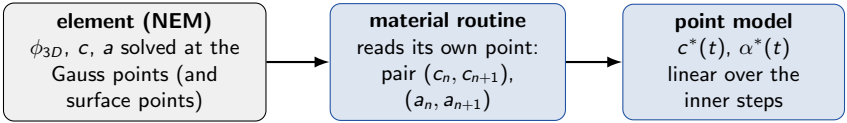
$\sum_i \bar{\phi}_i$ scales the consumption of c and a $\rightarrow$ c enters the front term of Eq. 34 $\rightarrow$ spreading of ϕ_{3D} $\rightarrow$ growth α (Eq. 36) $\rightarrow$ stress

Equations

$$
\begin{aligned}
c^*(t) &= c_0^*\left[\tfrac{t_{n+1}-t}{\Delta t}c_n + \tfrac{t-t_n}{\Delta t}c_{n+1}\right], \quad \alpha^* \text{ likewise} \\
0 &= d\,\nabla^2 c - g\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, c, \qquad \bar{\phi}_i = \phi_i\psi_i \\
0 &= d_a\nabla^2 a - g_a\,\phi_{3D}\,\textstyle\sum_i \bar{\phi}_i\, a \\
\dot{\phi}_{3D} &= \beta\nabla^2\phi_{3D} + k_\alpha\alpha - \|\nabla\phi_{3D}\|\,\tfrac{r\,c}{k+c}\,\mathbf{n}_{\nabla\phi_{3D}}\cdot\mathbf{n}_{\nabla c}, \qquad \dot{\alpha} = k_\alpha\phi_{3D}
\end{aligned}
$$

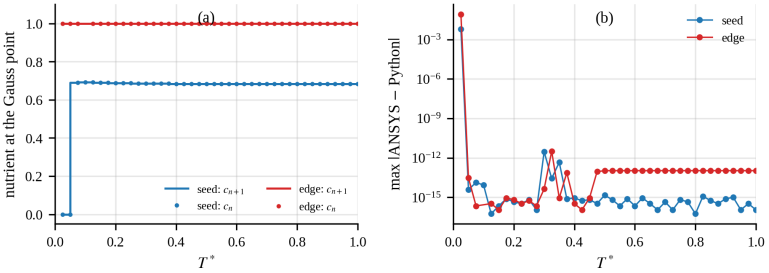
$g\phi$ in Eq. 35 of the paper is a misprint; $g\phi_{3D}c$ is used. ϕ_{3D}: field, ϕ_i: point model.

Nutrient and antibiotic at the Gauss points



- No interpolation from nodes: the fields live at the integration points
- The antibiotic in the second nutrient field takes the same path
- One increment behind: the fields are solved after the material routine
- A short first load step solves the fields before the point model starts

First ANSYS run: what is checked



- (a) c_{n+1} of one increment is c_n of the next: the pair is passed on without a jump
- (b) point model in ANSYS = Python with the same inputs: 2×10^{-12} or better in every step (four runs, with and without antibiotic)
- The first step needs up to 38 Newton iterations from the start state; with a tolerance-controlled Newton (up to 50) both codes agree there too

Time: one clock for both models

- $T^* = 1$ = duration of the point model's data
- s follows from it, not fitted (synthetic: $s = 0.1$)
- Check (synthetic): error 7×10^{-7}; with scaling 0.019

Examples and checks

Examples (synthetic, two species)

- A centre, B edges
- A and B mixed
- with / without antibiotic

Checks

- equals point model alone
- ANSYS = Python
- $0 \leq \phi \leq 1$
- ϕ_{min}, step, mesh

Results to follow

Assumptions and open points

- Assumptions: linear interpolation, $\sum_i \bar{\phi}_i$ one step late, start values of new points, ϕ_{min}
- No transport of species between points
- Front term needs upwinding
- Antibiotic: synthetic data only ($b = 0$ in the calibration)

Appendix: notation

ϕ_{3D}, α	biofilm phase field (ϕ of Klempt et al. 2024) and growth ($\mathbf{F}_g = \alpha\mathbf{I}$, $\alpha(0) = 1$), Klempt et al. 2024
c, a	nutrient and antibiotic fields (the element's first and second nutrient)
d, g, d_a, g_a	diffusion and consumption of nutrient and antibiotic
β, k_α, r, k	diffusion of ϕ, growth rate, front growth and Monod constant (Eq. 34, 36)
ϕ_i, ψ_i, ϕ_0	volume fraction and living fraction of species i, auxiliary void (Klempt et al. 2026)
c^*, α^*, b_i	nutrient and antibiotic of the point model, antibiotic sensitivity
$\bar{\phi}_i$	living volume fraction of species i, $\phi_i\psi_i$ (Klempt et al. 2026)
T^*, s	normalised time; point-model time per unit T^*